

Adaptive FeFET-Based LIF Neuron for Regression Tasks on Spiking Neural Networks

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Abstract—Spiking neural networks (SNNs) offer event-driven, energy-efficient computation for neuromorphic hardware, and in regression tasks, enable sparse, power-efficient processing of continuous data without compromising accuracy. However, most existing SNN implementations are constrained by fixed neuron parameters such as the membrane time constant (τ_{mem}) and firing threshold (V_{th}). We present a compact FeFET-based leaky integrate-and-fire (LIF) neuron with intrinsic dual tunability of τ_{mem} and V_{th} . The design exploits ferroelectric polarization states for real-time adaptation and employs a series FeFET–NMOS non-linear voltage-divider driving a CMOS inverter, thereby eliminating the need for bulky comparators and reducing circuit complexity. We demonstrate the proposed neuron’s effectiveness by training an SNN for regression of NMOS I_D – V_{DS} characteristics from 16-nm node data. Results show that dual tunability significantly improves accuracy and efficiency over fixed-parameter networks, establishing the proposed dual-adaptive LIF neuron as a promising building block for adaptive neuromorphic hardware.

Index Terms—SNN, LIF Neuron, FeFETs, tunable-neuron dynamics.

I. INTRODUCTION

Neural networks have transformed problem-solving methodologies, reshaping the way complex systems are analysed and controlled. Sustaining this progress requires hardware platforms that can inherently support adaptive, learning-centric paradigms and exploit event-driven computation. This has motivated the development of neuromorphic architectures that collocate memory and processing, drawing inspiration from the brain’s co-location of storage and computation, to overcome the memory-wall bottleneck and achieve energy-efficient information processing [1]. Among emerging neuromorphic paradigms, spiking neural networks (SNNs) have emerged as a preferred choice, offering a hardware-friendly alternative to conventional artificial neural networks (ANNs). Unlike ANNs, which operate continuously on input signals, SNNs operate in an event-driven manner, encoding information in discrete spikes and propagating events only when changes occur. SNNs

naturally exploit temporal processing and sparse activity, offering significant advantages in latency and energy efficiency, making them especially suited for hardware realization [2]. Moreover, their temporal processing capability makes SNNs a natural choice for time-series and sensory signal processing.

Several hardware platforms have been explored to implement SNNs. FPGA-based realizations offer flexibility and rapid prototyping of large networks but suffer from high power consumption and area overhead, limiting large-scale deployment. Custom CMOS circuits, particularly analog and mixed-signal designs, have demonstrated compact neurons using capacitors, resistive elements, and comparators, but face challenges of circuit complexity and scalability. More recently, approaches using emerging non-volatile devices such as memristors and ferroelectric field-effect transistors (FeFETs) have been explored to embed neuronal dynamics directly within the device [3], [4], thereby drastically reducing circuit complexity but have fixed temporal dynamics post-fabrication. Specifically, FeFETs-based implementations offer a CMOS-compatible option that combines non-volatility, low-voltage operation, and programmable polarization states, making them particularly promising for adaptive spiking neuron design.

Regardless of the platform, SNN hardware ultimately relies on the implementation of the spiking neuron model, the abstraction that defines the computational core of SNNs. Among different SNN models proposed, the leaky integrate-and-fire (LIF) neuron has emerged as the most widely adopted model for hardware due to its simplicity, scalability, and ease of circuit realization. Analogous to a biological neuron, the behaviour of an LIF neuron is defined by two key parameters, *viz.*, the membrane time constant ($\tau_{mem} = RC$), which sets the integration window, and the firing threshold (V_{th}), which determines excitability. These parameters govern the membrane potential $U(t)$, as expressed in Eq. 1, where $I_{in}(t)$ is input driving current.

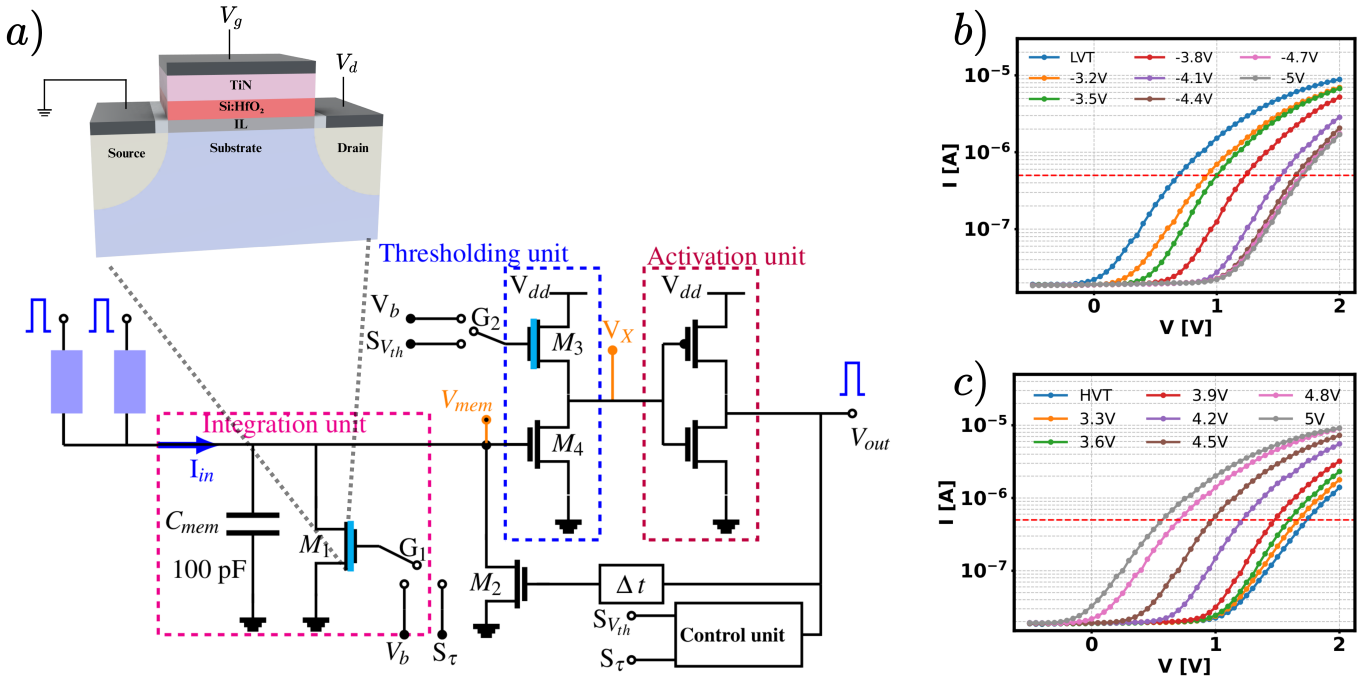


Fig. 1. (a) Schematic of the proposed LIF neuron, with the inset showing a typical FeFET device. Different resistance states of M_3 are obtained by modulating the stored threshold voltage (V_T) of the FeFET through voltage pulses of 500-ns width and varying amplitude. (b) FeFET resistance increases with increasing negative pulse amplitude, whereas (c) it decreases with increasing positive pulse amplitude. The threshold voltage V_T is extracted using the constant-current method, indicated by the dashed red line in (b) and (c). Data in (b) and (c) are adapted from [5].

$$\tau_{mem} \frac{dU(t)}{dt} = -U(t) + I_{in}(t)R \quad (1)$$

In a network of LIF neurons, these parameters, together with synaptic weights, define the overall network dynamics. Conventionally, SNN training optimizes only synaptic weights, treating τ_{mem} and V_{th} as fixed parameters. However, recent studies suggest that adaptivity in these neuron-intrinsic parameters, in addition to synaptic weight updates, can significantly enhance performance [6].

This has motivated research interest in hardware implementations supporting tunable τ_{mem} [7] and adaptive V_{th} [8]. Despite this, most of the reported hardware LIF neurons continue to suffer from limited post-fabrication tunability of intrinsic parameters [4]. Recent demonstrations using FeFETs and anti-ferroelectric FETs (AFEFETs) have shown promise in achieving tunable threshold control at the device level [3], [9]. However, existing approaches typically allow tuning of either τ_{mem} or V_{th} , not both simultaneously, and most evaluations focus on classification tasks rather than regression, where adaptability of both parameters is critical for capturing diverse patterns in training data.

In this work, we propose an adaptive FeFET-based LIF neuron that enables dual tunability of both τ_{mem} and V_{th} . The proposed design employs a simple series combination of an FeFET and an NMOS transistor, forming a non-linear voltage divider paired with a CMOS inverter, thereby reducing circuit complexity. By dynamically programming ferroelectric

polarization states, both the integration time constant and firing threshold can be precisely adjusted. We demonstrate the utility of this dual-adaptive neuron in regression tasks with SNNs, where parameter flexibility directly translates to improved accuracy and efficiency. Section II discusses the proposed neuron circuit and its design methodology, Section III reports regression performance benchmarks, and Section IV concludes the work. To the best of our knowledge, a CMOS-compatible FeFET-based LIF neuron with dual tunability in τ_{mem} and V_{th} has not been reported earlier, establishing a compact and energy-efficient LIF circuit for next-generation neuromorphic processors.

II. PROPOSED FEFET-BASED NEURON CIRCUIT

Fig. 1(a) illustrates the proposed FeFET-based LIF neuron with intrinsic tunability in both τ_{mem} and V_{th} . The architecture comprises three functional units, *viz.*, an integration unit for membrane potential accumulation, a thresholding unit for spike initiation, and an activation unit for output generation and reset. The FeFET devices used in the circuit serve as the primary source of tunability, enabling dual adaptation of τ_{mem} and V_{th} within a compact CMOS-compatible implementation. The following subsections first describe the FeFET device operation, then detail the functional units of the neuron, and finally present the tunability of neuronal parameters.

A. FeFET: A Tunable- V_{th} Device

The FeFET has emerged as a compelling device for neuromorphic computing following the discovery of ferroelec-

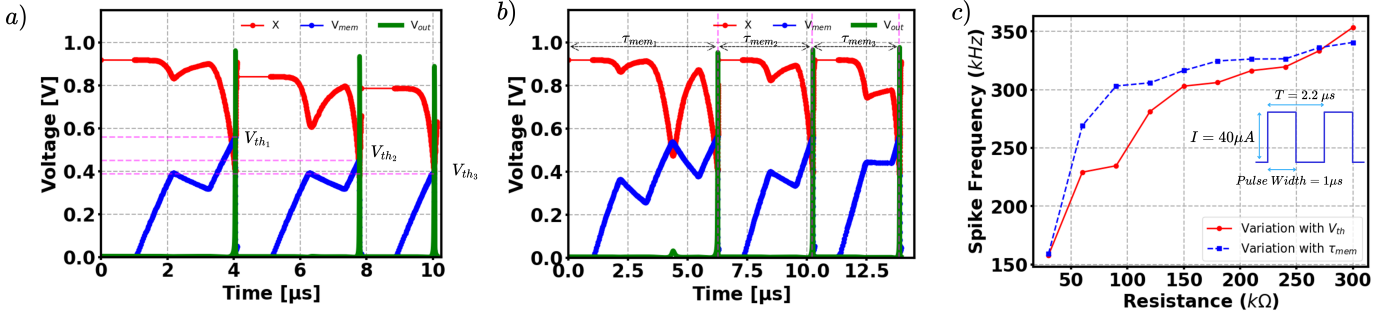


Fig. 2. SPICE simulation results demonstrating tunability of (a) V_{th} by programming the resistance of M_3 and (b) τ_{mem} by programming the resistance of M_1 . The evolution of V_{mem} , V_X , and V_{out} during neuron operation is also shown. (c) Modulation of spiking frequency through simultaneous variation of τ_{mem} and V_{th} . The periodic input current pulse used in the simulation is shown in the inset of (c).

tricity in CMOS-compatible hafnium oxide [10]. A hafnia-based ferroelectric layer embedded in the gate stack allows polarization-induced shifts in V_{th} of the FET. Because HfO_2 retains ferroelectricity at nanometre thickness and is fully CMOS-compatible, FeFETs enable dense, low-energy, in-memory computation. Furthermore, FeFETs can be operated as programmable resistances through V_{th} modulation driven by partial ferroelectric domain switching. Applying gate-voltage pulses of controlled amplitude and polarity gradually alters domain polarization states in the HfO_2 layer [11]. This results in graded, non-volatile conductance changes under a fixed read bias, effectively tuning device resistance. In the proposed neuron, this resistance-modulation capability is exploited to enable real-time adaptation of both τ_{mem} and V_{th} , thereby providing additional degrees of freedom for learning temporal patterns in training data.

B. Functional Units of the Neuron

The first functional block in the LIF circuit, shown in Fig.1(a) is the integration unit. It comprises a membrane capacitance (C_{mem}) and a FeFET transistor (M_1) acting as a variable resistor. During the spike integration phase, gate terminal G_1 is biased at V_b , setting $\tau_{mem} = M_1 \times C_{mem}$. The input current waveform is hence integrated as the capacitor voltage (V_{mem}) with a leakage set by τ_{mem} . This represents the ‘Leaky-Integrate’ part of the LIF neuron. Next is the thresholding unit, which consists of a series combination of a FeFET (M_3) and an NMOS transistor (M_4). When V_{mem} is below the threshold of M_4 , the intermediate node V_X remains high, keeping the activation stage inactive. Following this is the activation unit, implemented with a CMOS inverter. Once V_{mem} exceeds the threshold of M_4 , V_X drops, triggering an output spike (V_{out}). The spike event simultaneously discharges V_{mem} via the reset transistor (M_2) and activates the control unit, which generates the voltage signals used to adjust τ_{mem} and V_{th} . This corresponds to the ‘Fire’ operation of the LIF neuron.

C. Dual Tunability Mechanism

During the tuning phase, gate terminals G_1 and G_2 are driven by control signals $S_{\tau_{mem}}$ and $S_{V_{th}}$, respectively. These

signals modulate FeFET resistance, enabling real-time adaptation of τ_{mem} and V_{th} to minimize the network’s global training loss. The FeFET’s wide programmable resistance range is shown in Fig. 1(b) and 1(c). Fig. 1(b) shows measured I_D – V_G characteristics, illustrating the transition from a low- V_{th} (LVT, low-resistance) to a high- V_{th} (HVT, high-resistance) state, while Fig. 1(c) shows the reverse transition. The measurement data is adapted from [5], with fabrication and measurement details reported in [12].

The proposed neuron was validated through SPICE simulations using the 65-nm predictive technology model (PTM) [13]. A periodic input current waveform was applied, and the evolution of V_{mem} , V_{out} , and V_X was monitored, as shown in Figs. 2(a) and 2(b). Fig. 2(a) demonstrates V_{th} modulation by programming M_3 at three different resistance levels, while Fig. 2(b) illustrates τ_{mem} modulation by programming M_1 . Fig. 2(c) further shows that spike frequency can be tuned across a wide range through simultaneous modulation of V_{th} and τ_{mem} . In these simulations, the FeFET was modelled as a variable resistor.

III. APPLICATION MODELLING

The effectiveness of the proposed neuron was evaluated on the fundamental machine learning task of regression, a supervised learning problem in which the target variable is continuous. To adequately challenge the network’s learning capabilities, we adopt a non-linear regression task [14], which is not trivial due to the poor performance of linear models. Specifically, the task was defined such that the membrane potential of the output neuron should follow the output characteristics of an NMOS transistor, i.e., predicting I_D for a given V_{GS} and V_{DS} . The training data consists of I_D – V_{DS} curves generated using SPICE simulations with the 16-nm PTM model [13]. For each simulation, V_{DS} is swept across multiple V_{GS} values to obtain corresponding I_D values. Each training sample thus captures the evolution of I_D with V_{DS} at a fixed V_{GS} , providing a realistic dataset for benchmarking the proposed neuron network.

The SNN architecture employed for this task is shown in Fig. 3(a). It consists of two input neurons, a hidden layer of 256 neurons, and one output neuron. The two inputs

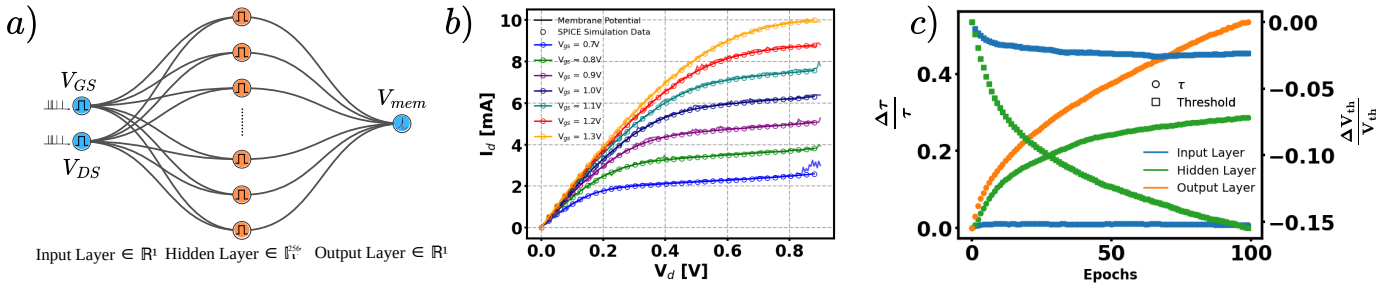


Fig. 3. (a) SNN architecture employing the proposed FeFET-based LIF neuron for predicting NMOS output characteristics. (b) Predicted I_D - V_{DS} characteristics from the SNN compared with SPICE simulation data. (c) Evolution of τ_{mem} and V_{th} across network layers during training, demonstrating parameter adaptation.

correspond to V_{DS} and V_{GS} , while the output neuron's membrane potential is trained to predict I_D . All neurons were implemented as LIF neurons, with the output neuron configured not to fire. Importantly, all neurons have learnable τ_{mem} and threshold values, enabled by the proposed circuit in Section II, thereby providing additional degrees of freedom during training. Training was performed using back-propagation through time (BPTT), a standard method for training SNNs [2], with Mean Squared Error (MSE) as the loss function between the predicted membrane potential and the target I_D . This enables the network to converge towards an optimal solution. Fig. 3(b) shows excellent agreement between predicted and actual I_D values from the PTM dataset, validating the network's regression capability. Furthermore, Fig. 3(c) illustrates the evolution of τ_{mem} and V_{th} across different layers during training. The observed parameter drift highlights the necessity for tunable and adaptive neuronal parameters, thus demonstrating that fixed τ_{mem} and V_{th} would significantly impair network learning performance.

IV. CONCLUSION

In this work, we demonstrated a compact FeFET-based LIF neuron with dual tunability of both τ_{mem} and V_{th} . By exploiting ferroelectric polarization states, the proposed design enables real-time parameter adaptation and eliminates bulky comparators by employing a CMOS inverter in the activation unit. Validation on a non-linear regression task highlights that tunability is essential not only for accurate processing of diverse temporal signals but also for enabling novel applications, such as data-driven modelling of semiconductor devices. To the best of our knowledge, this is the first CMOS-compatible demonstration of a dual-adaptive LIF neuron, positioning FeFETs as a promising device-level building block for adaptive neuromorphic hardware.

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